Apurbo Barua

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Education

University of Arizona — Tucson, AZ
B.S. in Computer Science, Minor in Business Administration | GPA: 3.51/4.00 — *Aug 2022 – May 2026*
- **Honors:** Dean's List, Academic Distinction, Wildcat Excellence Scholarship
- **Leadership:** Chairperson of Academic Excellence, Delta Chi Fraternity

Technical Skills

AI/ML: PyTorch, TensorFlow, scikit-learn, XGBoost, Meta Prophet, Hugging Face, LangChain, RAG, LLMs, Pandas
Languages: Python, Java, TypeScript, JavaScript, SQL
Backend & APIs: FastAPI, Node.js, Express.js, Django, REST APIs, GraphQL, Microservices
Frontend: Streamlit, React, HTML, CSS, Tailwind CSS
Databases & Cloud: PostgreSQL, MongoDB, DynamoDB, Redis, AWS (Lambda, API Gateway, EC2, S3, IAM), Azure
DevOps & Systems: Docker, Kubernetes, KEDA, Git, GitHub Actions (CI/CD), Linux/Unix, CloudWatch

Professional Experience

Software Engineer Intern — Jan 2026 – May 2026
PwC — *Tucson, AZ*
- Built an agentic AI chatbot using Claude 3 Haiku via AWS Bedrock with a dual-pass engine that generates and executes live SQL, enabling stakeholders to query workforce data in plain English.
- Developed an XGBoost model with SHAP explainability to predict project extension risk, surfacing operational drivers behind delivery delays for proactive staffing decisions.
- Architected a full-stack workforce planning platform (React, TypeScript, Node.js, AWS RDS PostgreSQL) consolidating 5 enterprise systems into a unified dashboard, deployed on AWS EC2.

Software Engineer Intern — Aug 2025 – Dec 2025
EduTrend (Quanticore LLC) — *Remote*
- Designed GraphQL schemas, resolvers, and MongoDB service layers for admin management in a TypeScript mobile backend.
- Built secure authentication and role-based access control using Clerk, supporting OAuth-based login flows for mobile clients.
- Diagnosed and resolved GitHub Actions CI/CD pipeline failures by fixing cross-platform ESLint and Linux build issues, restoring automated testing and deployment.

Undergraduate Teaching Assistant (CS 110) — Jan 2024 – May 2026
University of Arizona, Dept. of Computer Science — *Tucson, AZ*
- Built Python automation scripts for grading and testing, reducing manual evaluation time by 10+ hours per week.
- Taught core programming, data structures, and algorithms to 150+ students per semester, debugging Python and Java code in labs and office hours.

Projects

AgriSense github.com/ApurboBarua17/AgriSense
- Built a crop-disease detection app using HuggingFace, PyTorch, and GPT-4o-mini that identifies plant diseases from a photo and generates a USDA-backed treatment plan for farmers.
- Deployed 5 FastAPI microservices on Kubernetes with Redis and Meta Prophet to serve real-time disease diagnosis and market price forecasts for crop sell-or-treat decisions.

UA CS Degree Planner github.com/ApurboBarua17/RAG_chatbot
- Engineered a RAG-powered academic advising app using FastAPI, Groq (Llama 3), and ChromaDB that bypasses weeks of advising wait times by instantly guiding students through specific CS course tracks.
- Implemented prompt-engineered guardrails and tested outputs with real students to actively reduce AI hallucinations; deployed via Vercel and Hugging Face Spaces.

Tucson Crime Analysis github.com/ApurboBarua17/tucson-crime-analysis
- Analyzed 170,000+ municipal records using Python, Pandas, and scikit-learn, proving that incidents actually peak during daytime hours near retail and park locations rather than at night.
- Trained Random Forest and Logistic Regression models to forecast high-risk spatial and temporal zones, providing actionable insights for optimal police surveillance scheduling.

ResellIQ github.com/ApurboBarua17/ResellIQ
- Architected a resale pricing tool using React, FastAPI, PostgreSQL, and Redis that pulls live eBay and Discogs listings to help resellers price items before listing.
- Integrated GPT-4o-mini to generate recommended asking prices and ready-to-paste listing descriptions from live marketplace data, deployed on a Kubernetes and Docker backend.

Certifications

IBM Generative AI Engineering – IBM / Coursera — 2025
AWS Fundamentals – Amazon Web Services / Coursera — 2025